This unit is assessed by means of a written examination paper of 1 hour 30 minutes duration. The paper will consist of objective, short-answer and long-answer questions. Students may be required to apply their knowledge and understanding of physics to situations that they have not encountered before. The total number of marks available for this examination paper is 80. It contributes 40% to IAS and 20% to the IAL in Physics.

It is recommended that students have access to a scientific calculator for this paper.

Students will be provided with the formulae sheet shown in *Appendix 5*. Any other physics formulae that are required will be stated in the question paper.

The quality of written communication will be assessed in the context of this unit through questions which are labelled with an asterisk (*). Students should take particular care with spelling, punctuation and grammar, as well as the clarity of expression, on these questions.

1.3 Mechanics

This topic leads on from the Key Stage 4 programme of study and covers rectilinear motion, forces, energy and power. It may be studied using applications that relate to mechanics, for example, sports.

Students will be assessed on their ability to:		Suggested experiments
1	use the equations for uniformly accelerated motion in one dimension, $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$	
2	demonstrate an understanding of how ICT can be used to collect data for, and display, displacement/time and velocity/time graphs for uniformly accelerated motion and compare this with traditional methods in terms of reliability and validity of data	Determine speed and acceleration, for example use light gates
3	identify and use the physical quantities derived from the slopes and areas of displacement/time and velocity/time graphs, including cases of non-uniform acceleration	
4	investigate, using primary data, recognise and make use of the independence of vertical and horizontal motion of a projectile moving freely under gravity	Strobe photography or video camera to analyse motion
5	distinguish between scalar and vector quantities and give examples of each	
6	resolve a vector into two components at right angles to each other by drawing and by calculation	
7	combine two coplanar vectors at any angle to each other by drawing, and at right angles to each other by calculation	
8	draw and interpret free-body force diagrams to represent forces on a particle or on an extended but rigid body, using the concept of *centre of gravity* of an extended body	Find the *centre of gravity* of an irregular rod
9	investigate, by collecting primary data, and use $\Sigma F = ma$ in situations where m is constant (Newton's first law of motion ($a = 0$) and second law of motion)	Use an air track to investigate factors affecting acceleration